THE CENTER OF A FINITE SIMPLE ALGEBRA IS A FINITE FIELD EXTENSION

THE STACKS PROJECT AUTHORS

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Editorial note. The proof is the author's complete argument; both steps name their input and no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Wedderburn's theorem. Same chapter, Theorem theorem-wedderburn. Supplies the description of the algebra as a matrix algebra over a skew field finite over the ground field.
- Centralizers commute with tensor products. Same chapter, Lemma lemma-centralizer. Supplies that the center of the matrix algebra is the tensor product of the ground field with the center of the skew field, which identifies the center.

Source reference. Original theorem 0747: sources/stacks-brauer/pinned-original-source.tex, source lines 358–358. Navigation pointer invoked at proof line 418; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0749: sources/stacks-brauer/pinned-original-source.tex, source lines 395–395. Navigation pointer invoked at proof line 420; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let A be a finite simple k-algebra. Then the center k' of A is a finite field extension of k.*

Date: stacks-project:brauer.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Write $A = \mathrm{Mat}(n \times n, K)$ for some skew field K finite over k, see Theorem 0747. By Lemma 0749 the center of A is $k \otimes_k k'$ where $k' \subset K$ is the center of K. Since the center of a skew field is a field, we win. $\square$